

Where Every Block Is — 2 libraries + 1 shortcut

MATLAB R2024b. Electrical & measurement blocks → **sps_lib**. Signal blocks (Scope, Pulse Generator, Product...) → **Simulink library**. If you remember part of the name → just type it on the canvas.

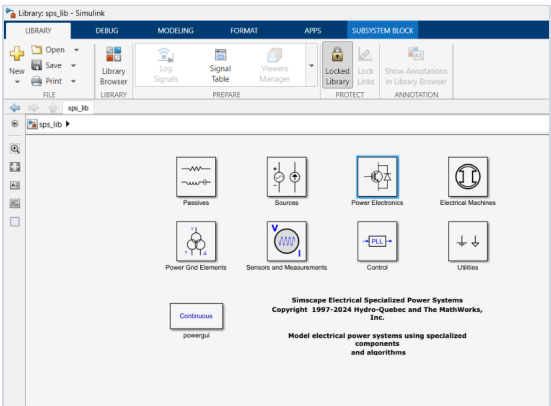
Method 1 — sps_lib (power blocks)

- 1. MATLAB command window → type `sps_lib` → Enter.
 - 2. Double-click a folder (Passives, Sources, Power Electronics...).
 - 3. **Drag** the block into your model window (keep both windows side by side).
 - 4. Back button (◀ at top-left) returns to the main page.
- powergui is on the main page itself.

Method 2 — Simulink library (signal blocks)

- 1. Command window → `open_system('simulink')` → Enter (or in your model: Simulation tab → **Library Browser** → **Simulink**).
- 2. Double-click a folder (Sources, Sinks, Math Operations...).
- 3. Drag the block into your model.

Shortcut (fastest): click empty canvas → type a keyword (last column below) → Enter.



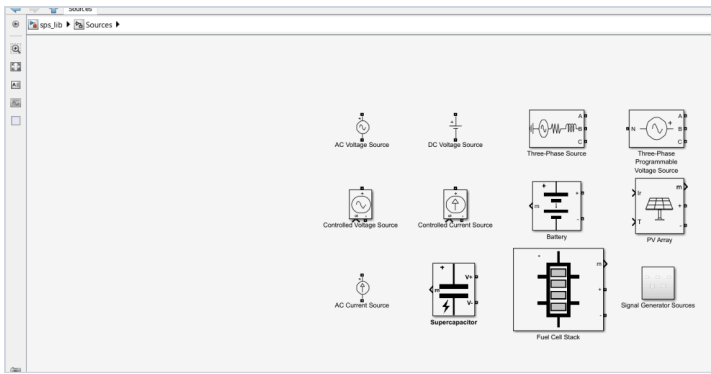
sps_lib main page (your screenshot): 8 folders + powergui

Method 1 · sps_lib — power & measurement blocks

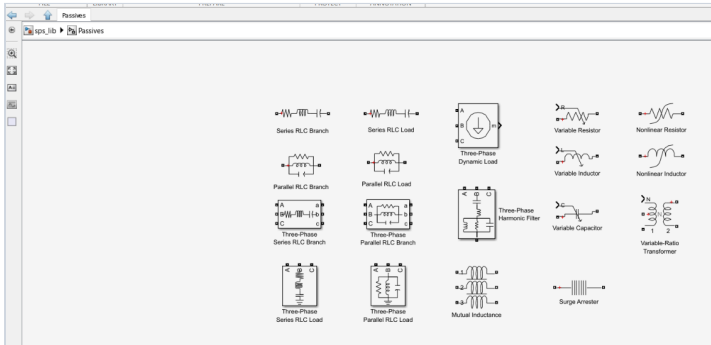
Block	sps_lib → folder	Used in	Type keyword
powergui	main page (bottom-left)	every experiment	powergui
AC Voltage Source	Sources	rectifiers	ac volt
DC Voltage Source	Sources	DC-DC, inverters	dc volt
Controlled Voltage Source	Sources	3.6, 4.3	controlled volt
Three-Phase Source	Sources	2.3b, 3.4	three-phase source
Sawtooth Generator	Sources → Signal Generator Sources (grey box, double-click) *	4.5	sawtooth
Series RLC Branch	Passives	every load, L, C	series rlc
Three-Phase Series RLC Branch	Passives	2.3b	three-phase series
Mutual Inductance	Passives	3.6	mutual
Linear Transformer	Power Grid Elements (not Passives!)	4.8, 4.9	linear trans
Diode	Power Electronics	rectifiers, buck	diode
Thyristor	Power Electronics	Exp-3	thyristor
Mosfet	Power Electronics	DC-DC, inverters	mosfet
IGBT/Diode (alternative switch)	Power Electronics	inverters (optional)	igbt
Universal Bridge	Power Electronics	2.7 (option), 3.4, 3.5	universal
PWM Generator (DC-DC)	Power Electronics → Power Electronics Control (grey box) *	4.6–4.9	pwm generator
Pulse Generator (Thyristor, 6-Pulse)	Power Electronics → Power Electronics Control *	3.4	6-pulse
PLL (3ph)	Control *	3.4	pll
Voltage Measurement	Sensors and Measurements → Sensors	everywhere	voltage meas
Current Measurement	Sensors and Measurements → Sensors	everywhere	current meas
Multimeter	Sensors and Measurements → Sensors	1.3, 2.7, 4.2...	multimeter
Three-Phase V-I Measurement	Sensors and Measurements → Sensors	2.3b, 3.4	v-i meas
Power / Power (3ph, Instantaneous)	Sensors and Measurements → Power	2.3	power

Mean (the plain one, <i>not</i> “Variable Frequency” / “Phasor”)	Sensors and Measurements → Miscellaneous	everywhere	mean
RMS	Sensors and Measurements → Miscellaneous	everywhere	rms
Fourier	Sensors and Measurements → Miscellaneous	2.6, 5.1	fourier
THD	Sensors and Measurements → Miscellaneous	5.2	thd
Ground	Utilities (the $\perp$ symbol, no label)	2.7, 2.8	ground

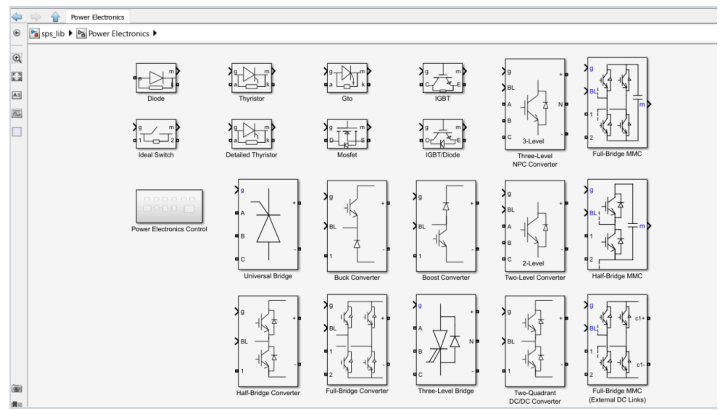
* Folder not visible in your screenshots — if it isn't there, just type the keyword on the canvas (always works).



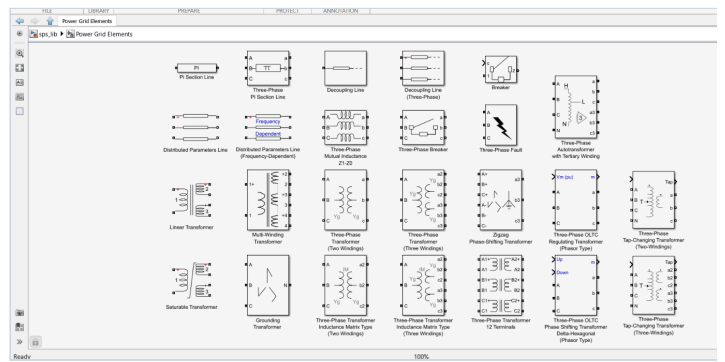
Sources



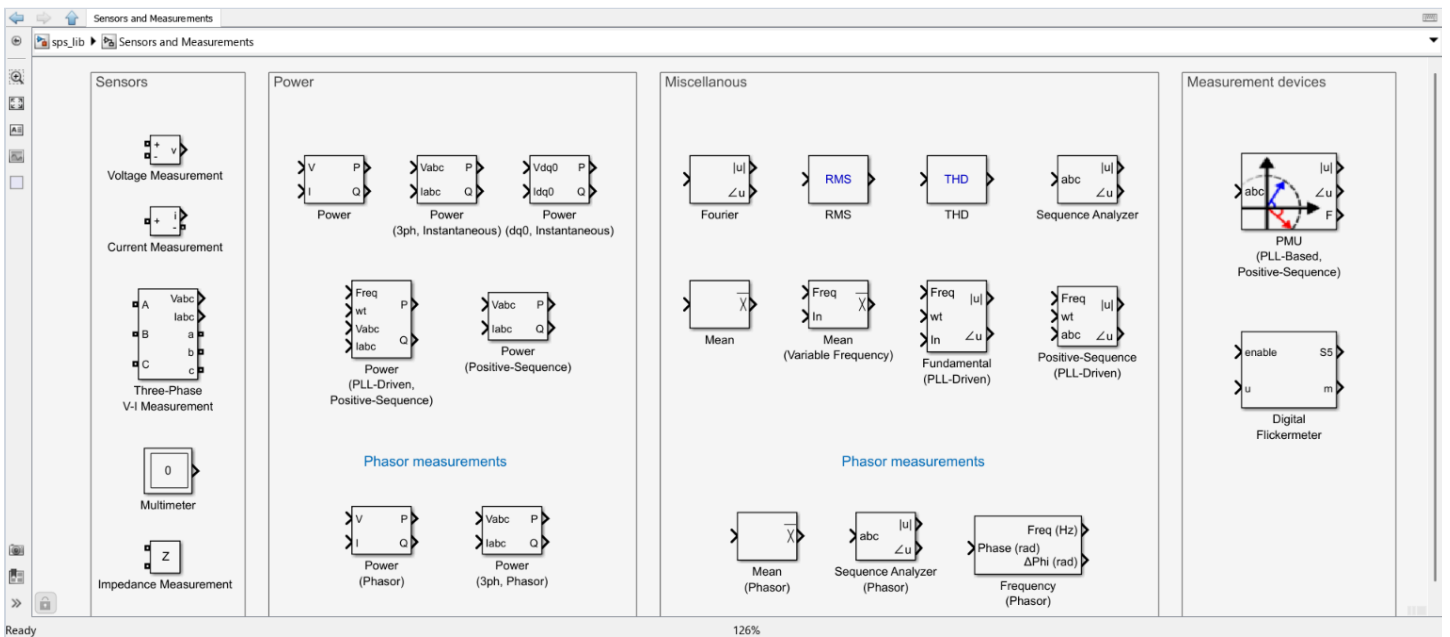
Passives



Power Electronics (Power Electronics Control = grey box)



Power Grid Elements → Linear Transformer (left)



Sensors and Measurements — Sensors (left), Power, Miscellaneous (Fourier, RMS, THD, Mean)

Method 2 · Simulink library — signal blocks (NOT in sps_lib)

These are ordinary Simulink blocks, so they are missing from sps_lib. Open them with `open_system('simulink')` or Library Browser → **Simulink**, or type the keyword.

Block	Simulink → folder	Used in	Type keyword
Scope	Sinks (also Commonly Used Blocks)	everywhere	scope
Display	Sinks	everywhere	display
Terminator	Sinks	unused m-ports	terminator
Pulse Generator	Sources	SCR gate, buck gate	pulse generator
Step	Sources	3.6, 4.3, 4.4, 4.7	step
Constant	Sources	α , D, 0, 1	constant
Sine Wave	Sources	inverter references	sine wave
Repeating Sequence	Sources	inverter carrier	repeating seq
Product	Math Operations	power $P = v \cdot i$	product
Divide	Math Operations	PF = P/S, efficiency	divide
Gain	Math Operations	$\times 3$, $\times 0.5$, $\times -1$, $\times 100$	gain
Sum	Math Operations	error, V_{AB}	sum
Mux / Demux	Signal Routing	multimeter, m-port	mux / demux
Goto / From	Signal Routing	2.1, gates [A] [B]	goto / from
Relational Operator	Logic and Bit Operations	PWM ($\geq$)	relational
Logical Operator	Logic and Bit Operations	NOT for lower switch	logical
Saturation	Discontinuities	duty limit 0...1	saturation
Transport Delay	Continuous	2nd SCR (3.2, 3.3)	transport
PID Controller	Continuous	4.7	pid
Data Type Conversion	Signal Attributes	only if gate data-type error	data type conv
Subsystem (Ctrl+G is easier)	Ports & Subsystems	2.2, 5.5	subsystem
Connection Port	Library Browser → Simscape → Utilities	5.5 only	connection port

Picking the right one when typing (Quick Insert)

Keyword	Several results appear → choose
mean	Mean with path "...Specialized Power Systems/Sensors and Measurements" (has <i>Fundamental frequency</i> box). Not "Mean (Variable Frequency)", not DSP "Mean".
diode / mosfet / thyristor	the one with path Specialized Power Systems/Power Electronics (Simscape Semiconductors versions won't connect).
pulse generator	plain Pulse Generator (Simulink/Sources) for single SCR / buck; (Thyristor, 6-Pulse) only for 3- ϕ bridge.
ac volt	AC Voltage Source (Specialized Power Systems/Sources).
scope	plain Scope (Simulink/Sinks) — not "Floating Scope", not "Spectrum Analyzer".

Exam routine: keep two windows open — your model + sps_lib . Power blocks → drag from sps_lib. Scope, Display, Pulse Generator, Product... → type the keyword on the canvas. If the keyword gives nothing → Library Browser → Simulink.